

Engineering Statistics

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Chapter 6: Sampling Distribution

Outline

- ▶ Sampling distributions
- ▶ Central Limit Theorem

Sample Variation

Suppose a random sample of size $n = 5$ is taken (without replacement) from the following population:

5, 8, 10, 10, 25, 26, 31, 31, 35, 37.

- Population mean:

$$\mu = 21.8.$$

- Population standard deviation:

$$\sigma = 12.26.$$

Sample Variation

- ▶ Sample 1: {5, 10, 26, 31, 35}

$$\bar{x} = 21.4, \quad s = 13.202.$$

- ▶ Sample 2: {5, 8, 25, 31, 37}

$$\bar{x} = 21.2, \quad s = 14.114.$$

- ▶ Sample 3: {10, 10, 31, 35, 37}

$$\bar{x} = 24.6, \quad s = 13.502.$$

Whenever we choose a random sample, we get different values of $\bar{x}$ and s .

Sampling Distribution

Sampling Distribution: An Illustration

A statistic, such as the sample mean $\bar{X}$, varies from sample to sample. The probability distribution of a statistic under repeated random sampling is called its **sampling distribution**.

Population

A finite population of size 10:

5	8	10	10	25
26	31	31	35	37

Population mean: $\mu = 21.8$

Population SD: $\sigma = 12.26$

Random sampling

Take a random sample of size $n = 5$ (without replacement).

Many possible samples

Different samples

Sample 1: {5, 10, 26, 31, 35}

$\bar{x} = 21.4, s = 13.202$

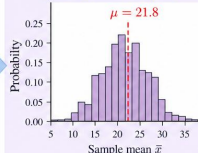
Sample 2: {5, 8, 25, 31, 37}

$\bar{x} = 21.2, s = 14.114$

Sample 3: {10, 10, 31, 35, 37}

$\bar{x} = 24.6, s = 13.502$

Sampling distribution of $\bar{X}$ (for all possible samples)



Why is the exact distribution difficult to find?

- Even in this small example (population size $N = 10$, sample size $n = 5$), there are $\binom{10}{5} = 252$ possible samples.
- To find the exact sampling distribution of $\bar{X}$, we would need to compute the sample mean for all 252 samples and then determine the probability of each value.
- For realistic data, the population size N is usually very large or infinite, and the number of possible samples is enormous (or infinite), making the exact distribution infeasible to obtain.

In practice

- We often use theoretical results, such as the Central Limit Theorem (CLT), to approximate the sampling distribution.
- For example, for large n ,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right),$$

even when the population distribution is not normal.

Point Estimation

- ▶ A **statistic** is a function of the random variables in a random sample.
 - ▶ Therefore, **a statistic is also a random variable.**
- ▶ A statistic can be used to estimate an unknown population parameter.
 - ▶ We use $\bar{X}$ to estimate μ .
 - ▶ We use S^2 to estimate σ^2 .

Point Estimation

Let $X_1, \dots, X_n$ be a random sample of size n from a probability distribution $f(x)$, characterized by an unknown parameter θ .

- ▶ The statistic

$$\hat{\Theta} = h(X_1, \dots, X_n)$$

is called a **point estimator** of θ .

- ▶ $\bar{X}$ is a point estimator of μ .
- ▶ After the sample is observed, $\hat{\Theta}$ takes a particular numerical value $\hat{\theta}$, called a **point estimate** of θ .
 - ▶ $\bar{x}$ is a point estimate of μ .

Sampling Distribution

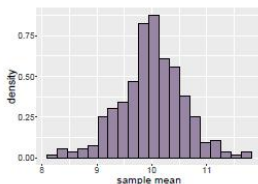
The probability distribution of a statistic under random sampling is called its **sampling distribution**.

- ▶ Statistical inference is possible because a statistic has a predictable probability distribution.
- ▶ A sampling distribution allows us to quantify the variability of a statistic from sample to sample.
 - ▶ How much can a statistic differ from the population parameter?
 - ▶ What amount of variability can occur simply due to random sampling?

Sampling Distribution

Q. How can we obtain a sampling distribution?

1. Generate a random sample of size n and compute the statistic of interest, such as $\bar{x}$.
2. Repeat this process many, many times.
3. Create a histogram of the computed statistics to approximate their sampling distribution.



Q. Can we actually do this **in practice** every time we collect data?

Sampling Distribution of $\bar{X}$

Let

$$X_1, X_2, \dots, X_n$$

be a **simple random sample** from a population with mean μ and variance σ^2 .

- ▶ We assume

$$X_1, \dots, X_n$$

are independent and identically distributed (**iid**).

- ▶ The sample mean is

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}.$$

- ▶ Because $X_1, \dots, X_n$ are random variables, $\bar{X}$ is also a random variable.

Mean and Variance of $\bar{X}$

If $X_1, X_2, \dots, X_n$ are iid with population mean μ and variance σ^2 , then

$$E(\bar{X}) = E\left(\frac{1}{n}X_1 + \dots + \frac{1}{n}X_n\right) = \mu.$$

Thus, $\bar{X}$ is an **unbiased estimator** of μ .

Also,

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n}X_1 + \dots + \frac{1}{n}X_n\right) = \frac{\sigma^2}{n}.$$

Therefore,

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}.$$

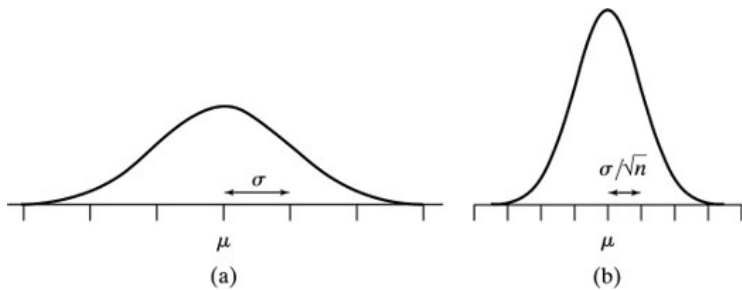
Standard Error of $\bar{X}$

- ▶ The standard deviation of a sampling distribution measures how much a statistic varies from sample to sample.
- ▶ The standard deviation of a sampling distribution is called the **standard error**.
- ▶ For the sample mean,

$$SE(\bar{X}) = \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

- ▶ Smaller σ gives a smaller standard error.
- ▶ Larger n gives a smaller standard error.
- ▶ Increasing n therefore improves the precision of $\bar{X}$ as an estimate of μ .

Standard Error of $\bar{X}$



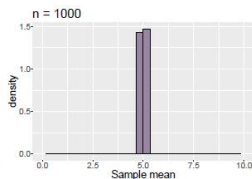
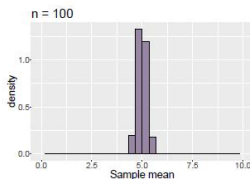
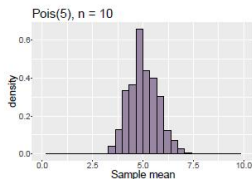
- ▶ The sampling distribution is centered at μ .
- ▶ The variability of $\bar{X}$ is smaller than the variability of X .

The Law of Large Numbers

The Law of Large Numbers (LLN):

As the sample size n increases, the sample mean $\bar{X}$ gets closer and closer to the population mean μ .

$$\bar{X} \rightarrow \mu$$



Shape of the Sampling Distribution of $\bar{X}$

So far, it looks like the sampling distribution of $\bar{X}$ is Normal. Is this always the case?

Normally distributed population:

If

$$X_i \stackrel{iid}{\sim} N(\mu, \sigma^2),$$

then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

for **any sample size** n .

Q. What if the X_i 's are **not** Normally distributed?

Central Limit Theorem

Central Limit Theorem (CLT):

Suppose $X_1, X_2, \dots, X_n$ are randomly selected from a population with mean μ and standard deviation σ .

If n is sufficiently large, then the sampling distribution of $\bar{X}$ is approximately Normal:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

even when the population itself is **not Normal**.

Central Limit Theorem

If n is sufficiently large,

$$\bar{X} \stackrel{\sim}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right).$$

Therefore,

- ▶ the population distribution does not need to be Normal;
- ▶ the sampling distribution of $\bar{X}$ becomes approximately Normal;
- ▶ many statistical inference procedures can therefore use Normal-based methods.
- ▶ Another way to write it:

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1), n \rightarrow \infty.$$

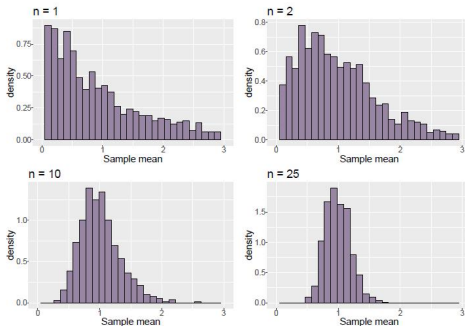
How Large a Sample Size?

It depends on the shape of the population distribution.

- ▶ If the population is Normal, the sampling distribution of $\bar{X}$ is Normal for any n .
- ▶ If the population is approximately Normal, relatively small samples may be sufficient.
- ▶ If the population is strongly skewed or otherwise far from Normal, a larger sample size is needed.
- ▶ A commonly used rule of thumb is $n \geq 30$, but the required sample size depends on the population distribution.

Central Limit Theorem

Suppose $X_i \stackrel{iid}{\sim} \text{Exp}(1)$. The population distribution is strongly right-skewed.

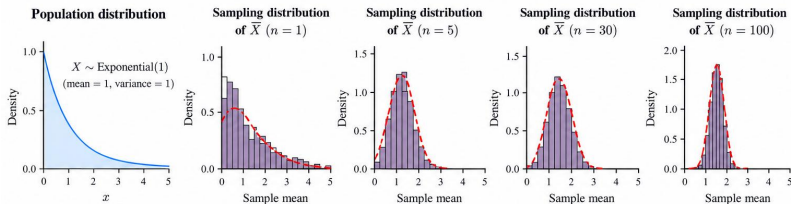


As n increases, the sampling distribution of $\bar{X}$ becomes increasingly Normal.

CLT Summary

Central Limit Theorem (CLT)

Even when the population distribution is not normal, the sampling distribution of the sample mean becomes approximately normal as the sample size n increases.



Central Limit Theorem

If $X_1, \dots, X_n$ are independent and identically distributed with mean μ and standard deviation σ , then

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \longrightarrow N(0, 1) \quad \text{as } n \rightarrow \infty.$$

Thus, for sufficiently large n , $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$, even when the population distribution is not normal.

Sampling Distribution of the Sample Proportion

Suppose

$$X_i \stackrel{iid}{\sim} \text{Ber}(p), \quad i = 1, \dots, n.$$

The sample proportion is

$$\hat{p} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Because

$$\sum_{i=1}^n X_i \sim \text{Bin}(n, p),$$

the sample proportion is a statistic whose value varies from sample to sample.

Sampling Distribution of the Sample Proportion

Let us look at an application of CLT. For the sample proportion,

$$E(\hat{p}) = p, \quad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

When n is sufficiently large and p is not too close to 0 or 1, the Central Limit Theorem gives

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

A commonly used condition is

$$np \geq 10 \quad \text{and} \quad n(1-p) \geq 10.$$

Re-cap

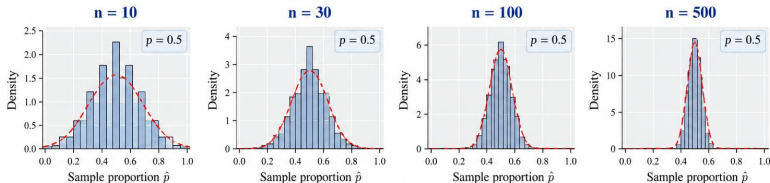
Sampling Distribution of a Proportion (Using CLT)

Let $X_1, \dots, X_n$ be independent Bernoulli(p) random variables, and let $\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample proportion.

For large n , by the Central Limit Theorem,

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \rightarrow N(0, 1),$$

so $\hat{p}$ is approximately $N\left(p, \frac{p(1-p)}{n}\right)$.



Conclusion

The sampling distribution of $\hat{p}$ is approximately normal for sufficiently large n , with mean p and standard error $\sqrt{p(1-p)/n}$.

Chapter Summary

- ▶ **Statistics vary from sample to sample.** Their probability distributions are called **sampling distributions**.
- ▶ **Larger samples give more precise estimates.** For the sample mean, $E(\bar{X}) = \mu$, $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$.
- ▶ **Normal population $\Rightarrow$ Normal sample mean.** If X is Normal,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

- ▶ **For non-Normal populations, use the CLT.** For sufficiently large n , the sampling distribution of $\bar{X}$ is approximately Normal.
- ▶ **Sample proportion is a special application of the CLT.** For sufficiently large n , $\hat{p}$ is approximately Normal.